

Kim Hanbyul 김한별

From biomedical engineering, semiconductors and medical AI to neuromorphic semiconductors — aiming to realize the principles of the brain as artificial-brain hardware.

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CURRENT POSITION

May 2026 – Jan 2027 (contract)

KIST · Research Intern — Neuromorphic Computing & Spiking Neural Networks

SNN modeling and dynamical analysis of cerebellar circuits; plasticity-control algorithms under neuromodulators

National R&D project · Participating researcher

Development of original cerebellum-inspired semiconductor technology integrated with non-volatile memory for real-time predictive control of physical AI systems

Next-Generation Intelligent Semiconductor Technology Development (Devices) — National R&D Program · RS-2026-25540328

PI: Dr. Jaewook Kim (KIST Center for Semiconductor Technology) · Participation May 1 – Dec 31, 2026 (8 months) · whole project Apr 2026 – Dec 2028

EDUCATION & TRAINING

Mar 2020 – Feb 2026 (graduated)

Konkuk University GLOCAL Campus — B.S.

College of Science and Technology — ICT Convergence Engineering / Biomedical Engineering

Minor: Department of System Semiconductor Engineering

Cumulative GPA 4.23 / 4.50 (percentile 97.3) · major 4.31 (98.1) · minor 4.38 (98.8) · 149 credits

Academic honors 5 semesters — 2020-1 · 2020-2 · 2023-2 · 2024-2 · 2025-1

2025

Medical AI — K-Digital Training (1,026 h)

Future Convergence Education Center, Daejeon · Konyang University Hospital Biomedical AI program (K-Digital Training, 1,026 h)

Key coursework

Biomedical engineering

Biosignal and System

Biomedical Signal Processing

Digital System for BME

Instrumentation

Fundamentals of Electronic Medical Devices and Experiments

Electronic Circuit Laboratory for BME

Semiconductor

Memory Semiconductor Devices

Semiconductor Equipment

Semiconductor Test and Packaging Process

Basic Semiconductor Test

Semiconductor Chemistry

Quantum Mechanics

Computing

Computer Architecture

Microprocessors & SOC

Computer Programming

SCHOLARSHIPS

2024-1 – 2025-2	★ National Scholarship for Science & Engineering — 2-year award, 4 semesters
2020-1	Admission-fee reduction scholarship
2020-1	Special academic support scholarship I (living costs) · II (tuition)
2020-2 · 2023-2	Contest scholarship (national innovation fund)
2023-1 · 2023-2	DreamPoint scholarship
2023-1	Competency development scholarship III (dormitory)

PROJECTS

2024 – 2025	AI assistant on ESP32-S3-BOX-3 (capstone) Built an IoT framework — Capstone Design 1 · 2
2026	ai-research-harness An AI workflow scaffold that keeps the researcher in charge — format roles, glossary, approval gates, error log brain-cdss-daejeon Brain clinical decision support system (Brain CDSS) — JavaScript / Django

SKILLS

Biomedical Engineering	Medicine	Electronics	C/C++	IoT	Biosignal processing
Semiconductor	RISC-V	Semiconductor process	Computer architecture (ALU · memory)	Logic operations	
Medical AI	PyTorch	Statistics	Swin UNETR (MRI)	Genomic data models	PACS Orthanc

ACTIVITIES

2023-2	Overseas field program — New Frontier
2024-2	Overseas cultural program (GLOCAL)

MILITARY SERVICE

2021 – 2022	Military service — armor crewman · discharged
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